

	Area	Dynamic Power	Max Frequency	LUTs	FFs
Vanilla	12,897 LEs (9%)	25.69 mW	132.21 MHz	12,897	8,514

Project Navigator

Hierarchy

Entity:Instance

Cyclone IV GX: EP4CGX150DF31I7AD

corelet

Tasks

Compilation

Task

Compile Design

Analysis & Synthesis

Fitter (Place & Route)

Assembler (Generate program)

Timing Analysis

EDA Netlist Writer

Edit Settings

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Power Analyzer Tool

Flow Summary

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Flow Status

Successful - Sat Dec 13 00:36:52 2025

Quartus Prime Version

25.1std.0 Build 1129 10/21/2025 SC Lite Edition

Revision Name

corelet

Top-level Entity Name

corelet

Family

Cyclone IV GX

Device

EP4CGX150DF31I7AD

Timing Models

Final

Total logic elements

12,897 / 149,760 (9 %)

Total registers

8514

Total pins

427 / 508 (84 %)

Total virtual pins

0

Total memory bits

0 / 6,635,520 (0 %)

Embedded Multiplier 9-bit elements

0 / 720 (0 %)

Total GXB Receiver Channel PCS

0 / 8 (0 %)

Total GXB Receiver Channel PMA

0 / 8 (0 %)

Total GXB Transmitter Channel PCS

0 / 8 (0 %)

Total GXB Transmitter Channel PMA

0 / 8 (0 %)

Total PLLs

0 / 8 (0 %)

All

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Type

ID

Message

223000 Starting Vectorless Power Activity Estimation

223001 Completed Vectorless Power Activity Estimation

218000 Using Advanced I/O Power to simulate I/O buffers with the specified board trace model

334003 Started post-fitting delay annotation

334004 Delay annotation completed successfully

215049 Average toggle rate for this design is 0.844 millions of transitions / sec

215031 Total thermal power estimate for the design is 320.36 mW

Quartus Prime Power Analyzer was successful. 0 errors, 4 warnings

corelet

Project Navigator Hierarchy

Entity: Instance

Cyclone IV GX: EP4CGX150DF31I7AD

corelet

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Slow 1200mV 100C Model Fmax Summary

<<Filter>>

	Fmax	Restricted Fmax	Clock Name	Note
1	132.21 MHz	132.21 MHz	clk	

This panel reports FMAX for every clock in the design, regardless of the user-specified paths where the source and destination registers or ports are driven by the same clock. Generated clocks, are ignored. For paths between a clock and its inversion, FMAX is scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Always use clock constraints and other clock reports for sign-off analysis.

All

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Type ID Message

- 223000 Starting Vectorless Power Activity Estimation
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corelet

Project Navigator Hierarchy

Entity: Instance

Cyclone IV GX: EP4CGX150DF31I7AD

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 - Thermal Power Dissipation
 - Core Dynamic Thermal

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Compilation

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- Compile Design
- Analysis & Synthesis
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- Timing Analysis
- EDA Netlist Writer
- Edit Settings

Power Analyzer Summary

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Power Analyzer Status	Successful - Sat Dec 13 00:36:52 2025
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	corelet
Top-level Entity Name	corelet
Family	Cyclone IV GX
Device	EP4CGX150DF31I7AD
Power Models	Final
Total Thermal Power Dissipation	320.36 mW
Core Dynamic Thermal Power Dissipation	25.69 mW
Core Static Thermal Power Dissipation	119.41 mW
I/O Thermal Power Dissipation	175.27 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data

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